

Vivek Hariharan

Baltimore, MD | hariharanviv@gmail.com | (203) 329-2744 | <https://github.com/hariharanviv>

SUMMARY

Senior PhD candidate in Biomedical Engineering with 8+ years of experience in molecular and cell engineering. Expertise in **lentiviral vector design, high-throughput screening (HTS), and multi-color intracellular flow cytometry**. Proven ability to **lead projects and mentor junior scientists**, with hands-on experience in molecular plasmid design, quantitative molecular biology (digital PCR), and NGS data analysis. Seeking an academic postdoctoral fellowship focused on viral latency, RNA virus engineering, and functional genomics.

EDUCATION

Johns Hopkins University Baltimore, MD
Ph.D. Candidate in Biomedical Engineering July 2020 – July 2026 (expected)
Dissertation: Development of a High-Throughput Lentiviral Screening Platform to Interrogate HIV-1 Neutralization Sensitivity and Mechanisms of Viral Interference (Dr. Robert Siliciano)

Georgia Institute of Technology Atlanta, GA
B.S. in Chemical and Biomolecular Engineering August 2016 – May 2020
Honors program
Thesis: Developing a Dengue and Zika Virus Vaccine by Glycan Engineering (Dr. Ravi Kane)

RESEARCH EXPERIENCE

Johns Hopkins University July 2020 – July 2026 (expected)
Dissertation Project 1: Development of high-throughput genetic screens (Dr. Siliciano)

- Developed a novel, scalable screening assay to quantify autologous antibody neutralization against diverse, patient-derived HIV-1 Env isolates (manuscript in preparation).
- Engineered and executed high-throughput screening assays for library-based functional genomics (e.g., antibody-Env interactions), establishing a platform adaptable for pooled CRISPR/sgRNA screens and lentiviral-based genetic engineering
- Utilized Bxb1 integrase for site-specific integration to generate stable cell lines, reducing recombination artifacts from lentivirus-based pooled genetic screens
- Executed the full genomic workflow for screen readouts: recovered barcodes from pooled populations, prepared NGS libraries via PCR, and analyzed sequencing data using custom bioinformatic pipeline
- Led, mentored, and trained a team of 3 graduate students in advanced molecular biology techniques, controlled experimental design, high-throughput assay development, and data analysis

Dissertation Project 2: Investigating superinfection in people living with HIV-1 (Dr. Simonetti and Dr. Siliciano)

- Led a study identifying, for the first time, defective interfering particles (DIPs) in people with HIV-1, uncovering a novel mechanism driving non-suppressible viraemia (***Nature Microbiology*, 2025**)
- Demonstrated that superinfection diversifies defective HIV-1 proviruses, thereby challenging the 'evolutionary dead end' paradigm and revealing a dynamic role for defective genomes in pathogenesis
- Characterized a novel, heterogeneous HIV-1 reservoir dominated by DIPs, distinct from typical clonally expanded cells
- Quantified and functionally validated patient-derived DIPs *in vitro* by developing and applying custom dPCR assays, site-directed mutagenesis, and multi-color reporter virus assays, providing the first proof of their ability to interfere with virus replication
- Verified the superinfection mechanism *in vivo* by isolating single, dually-infected patient cells (via p24 intracellular FACS followed by PCR), which also showed DIPs failed to prevent disease progression, offering critical insights for ongoing clinical trials

Georgia Institute of Technology December 2017 – May 2020
Undergraduate Researcher: Protein Engineering of Subunit Vaccines with Dr. Ravi Kane

- Engineered novel multivalent antigen presentation for increased vaccine efficacy by using VLPs
- Mastered core DNA and protein engineering techniques (PCR, mammalian/insect cell culture, protein purification, FPLC, ELISA)
- Managed 2 undergraduate students and trained new graduate students in molecular biology protocols
- Contributed to the development of vaccines that elicit broadly protective antibodies against Zika and Dengue virus

Harvard Medical School, Dana-Farber Cancer Institute

June 2019 – August 2019

Undergraduate Research: Harvard Stem Cell Institute Internship with Dr. Wayne Marasco

- Design of 'universal' Influenza vaccine by engineering viral surface proteins
- Design of large immunological flow cytometry panel to characterize mouse immune cells
- Attended Harvard Medical School Immunology summer course

Regeneron Pharmaceuticals

Intern in the Protein Biochemistry Department

June 2018 – August 2018

- Designed CRISPR-Cas9 plasmids for targeted gene editing in cancer cell lines
- Developed novel approach to characterizing monoclonal antibodies by size and charge

Intern in the Bioassay and Molecular Biology Department

May 2017 – August 2017

- Cultured and transfected multiple genetically engineered cell lines to support bioassay development
- Developed and optimized ELISA protocols for cytokine quantification

New York Medical College

June 2015 – August 2016

Research Student: Investigation of the enzyme ω -Amidase with Dr. Arthur J. L. Cooper

- Led research project in aberrant nitrogen metabolism, resulting in a **first-author publication** in *Biology*
- Independently designed and executed experiments, analyzed data, and wrote the manuscript
- Mastered core biochemical techniques (enzyme kinetics, spectrophotometry, HPLC)
- Found that ω -Amidase efficiently catalyzes the hydrolysis of oncometabolite 2-hydroxyglutarate
- Identified new substrates for ω -Amidase, expanding understanding of its physiological role, and uncovered previously unreported metabolic pathways

TEACHING EXPERIENCE

Johns Hopkins University

September 2022 – December 2022

Graduate Teaching Assistant (Course: Immunoengineering Principles and Applications)

- Developed and led hands-on laboratory experiments
- Managed all course grading for exams and final reports and held weekly office hours
- Led review sessions before exams
- Led paper discussions on primary literature

GRANTS & RESEARCH SUPPORT

G1. NIH F31 Ruth L. Kirschstein Predoctoral Fellowship (PI: **Vivek Hariharan**)

Submitted September 2024

Role: Principal Investigator

- Proposal successfully reviewed and discussed by NIH study section, ultimately not awarded
- Secured letters of support from Dr. Drew Weissman and Dr. Dan Barouch.

G2. NIH R01 NIAID P01AI169615 (PI: Robert F. Siliciano)

April 2022 – March 2027

Role: Graduate Research Assistant

- Support for dissertation research investigating HIV-1 pathogenesis, defective interfering particles (DIPs), and high-throughput neutralization screening.

CONFERENCE PROCEEDINGS

C1. **Vivek Hariharan**, J. Siliciano, R. Siliciano, F. Simonetti. (2026, March). "Superinfection with intact HIV-1 results in conditional replication of defective proviruses and nonsuppressible viremia in people living with HIV-1." HHMI Investigator Meeting, Chevy Chase, Maryland.

C2. M. Moskovljevic, **Vivek Hariharan**, K. Dhoshi, A. Zinsser, I. van Schermbeek, J. Lai, J. Siliciano, R. Siliciano. (2025, May). "Investigating Autologous Neutralizing Antibody Responses to HIV-1 in People Treated during Early Infection." HHMI Investigator Meeting, Janelia, Maryland.

C3. **Vivek Hariharan**, J. Siliciano, R. Siliciano, F. Simonetti. (2024, July). "Failure to Achieve Undetectable Viral Load caused by HIV Expression from Defective Proviruses and Large Reservoir Size." International AIDS Society: AIDS 2024, Munich, Germany. Oral Abstract.

C4. **Vivek Hariharan**, S. Murali, A. Arsiwala, R. Kane. (2019, April). "Designing Flavivirus Vaccines by Glycoengineering Protein Antigens." Georgia Institute of Technology Undergraduate Poster Symposium Poster, Orlando, Florida.

PUBLICATIONS

- P1. "Superinfection with intact HIV-1 results in conditional replication of defective proviruses and nonsuppressible viremia in people living with HIV-1", **Vivek Hariharan**, J. White, F. Dragoni, E. Fray, N. Pathoulas, M. Moskovljevic, H. Zhang, A. Singhal, J. Lai, S. Beg, E. Scully, E. Gilliams, D. Block, J. Keruly, R. Moore, J. Siliciano, F. Simonetti*, R. Siliciano*. *Nature Microbiology*, **2025**.
- P2. "Factors associated with resistance of HIV-1 reservoir viruses to neutralization by autologous IgG antibodies", N. McMyn, J. Varriale, H. Wu, **Vivek Hariharan**, M. Moskovljevic, T. Tan, J. Lai, A. Singhal, K. Lynn, K. Mounzer, P. Tebas, L. Montaner, R. Hoh, X. Yu, M. Lichterfeld, F. Simonetti, C. Kovacs, S. Deeks, J. Siliciano, R. Siliciano. *J Clin Invest*, **2025**.
- P3. "Provirus in CD4+ T cells reactive to autologous antigens contribute to nonsuppressible HIV-1 viremia," Wu, F., Moskovljevic, M., Dragoni, F., Jayaraman, S., Board, N. L., Camilo-Contreras, A., Bernal, S., **Vivek Hariharan**, Zhang, H., Lai, J., & others. *Science Translational Medicine*, **2025**.
- P4. "SIV proviruses seeded later in infection are harbored in short-lived CD4+ T cells", N. Sambaturu, E. Fray, **Vivek Hariharan**, F. Wu, C. Zitzmann, F. Simonetti, D. Barouch, J. Siliciano, R. Siliciano, R. Ribeiro, A. Perelson, C. Molina-Paris, T. Leitner. *Cell Reports*, **2025**.
- P5. "Tethered Antigenic Suppression Shields the Hemagglutinin Head Domain and Refocuses the Antibody Response to the Stalk Domain," Kim, D., Loeffler, K., Hu, Y., Arsiwala, A., Frey, S., Murali, S., **Vivek Hariharan**, Moreno, A., & Kane, R. S. *Chemistry*, **2025**.
- P6. "The latent reservoir of inducible, infectious HIV-1 does not decrease despite decades of antiretroviral therapy," McMyn, N. F., Varriale, J., Fray, E. J., Zitzmann, C., MacLeod, H., Lai, J., Singhal, A., Moskovljevic, M., Garcia, M. A., Lopez, B. M., **Vivek Hariharan**, & others. *The Journal of Clinical Investigation*, **2023**.
- P7. "Glycosylation as a Tool for Rational Vaccine Design", **Vivek Hariharan**, R. Kane. *Biotechnology and Bioengineering*, **2019**.
- P8. "Nanopatterning protein antigens to refocus the immune response," Arsiwala, A., Varner, C., McCaffery, J. N., Kell, A., Pendyala, G., Castro, A., **Vivek Hariharan**, Moreno, A., & Kane, R. S. *Nanoscale*, **2019**.
- P9. "The enzymology of 2-hydroxyglutarate, 2-hydroxyglutaramate and 2-hydroxysuccinamate and their relationship to oncometabolites," **Vivek Hariharan**, Denton, T. T., Paraszcszak, S., McEvoy, K., Jeitner, T. M., Krasnikov, B. F., & Cooper, A. J. L. *Biology*, **2017**.

AWARDS AND HONORS

- International AIDS Society (IAS) Young Investigator Award (Spring 2024)
- President's Undergraduate Award recipient (Spring 2019)
- Genentech's Outstanding Student Award runner up (Fall 2018)
- 1st place in Biochemistry and Grand Finalist at Westchester Science and Engineering (2016)
- Intel Science and Engineering Fair attendee (ISEF; 2016)

INTERESTS AND ACTIVITIES

- Mentor in the JHU BME Application Assistance program, aiding underrepresented students in pursuing BME PhD programs (2023 – present)
- JHU Basic Science Institute Summer Internship program mentor to undergraduate students exploring biomedical research at JHU (Summer 2024, 2025)
- Member of Hopkins Medical Orchestra (2025)